

CS 4412 - Data Mining

MOVIE & USER ARCHETYPES

MOVIELENS DATASET

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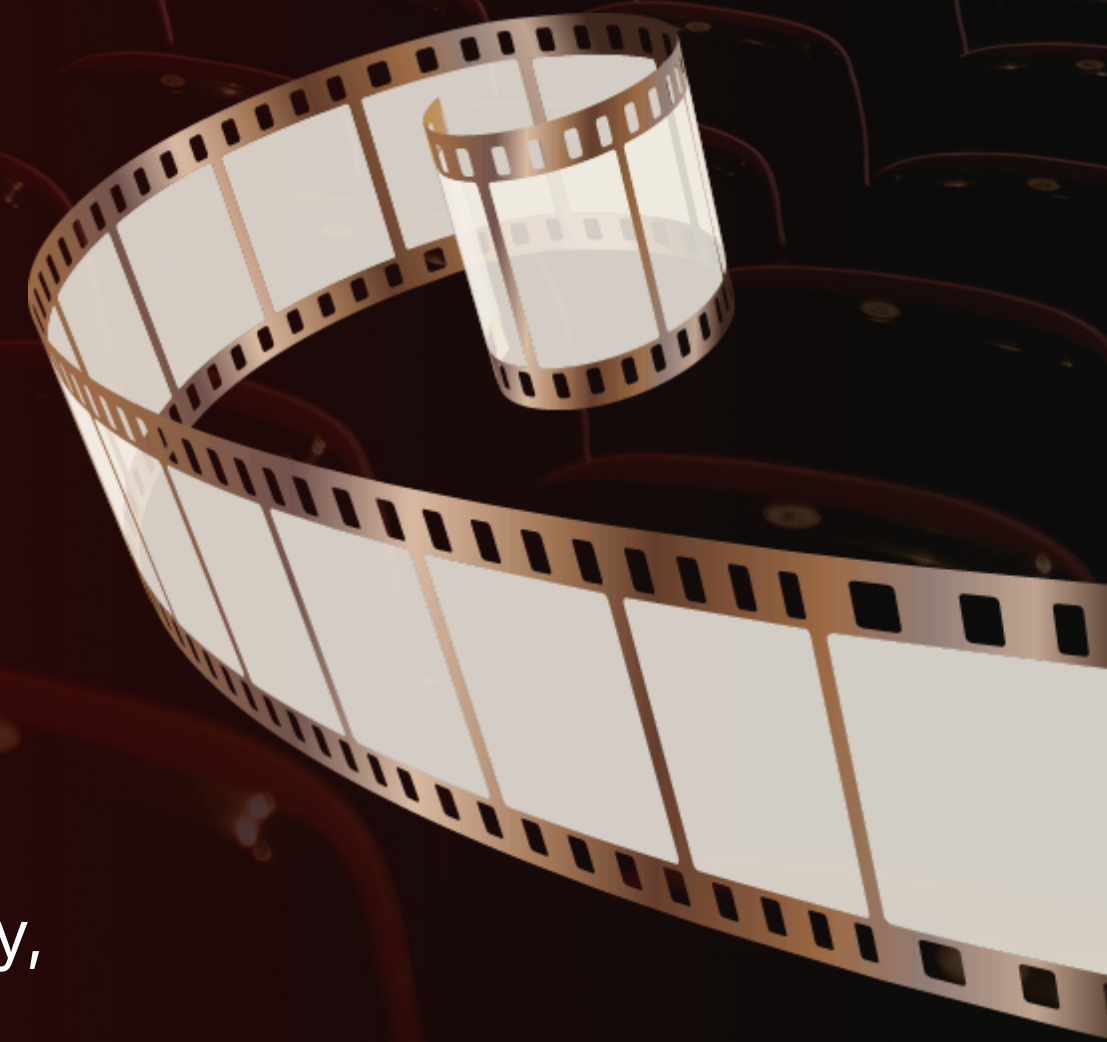
PROBLEM & QUESTIONS

Movies Lens Dataset:

- Over 87,000 movies with approximately 32 million ratings and 200,000 users
- Contains four csv files: movies, ratings, tags, and links

1. What natural groups of movies exist based on user ratings, genres, and tags?
2. How do user engagement patterns (rating volume and tagging) relate to rating behavior, and how do these factors form distinct user archetypes?
3. How do temporal patterns in rating activity (burst behavior, rating recency, and consistency over time) differ between user archetypes, and what anomalous rating patterns emerge?

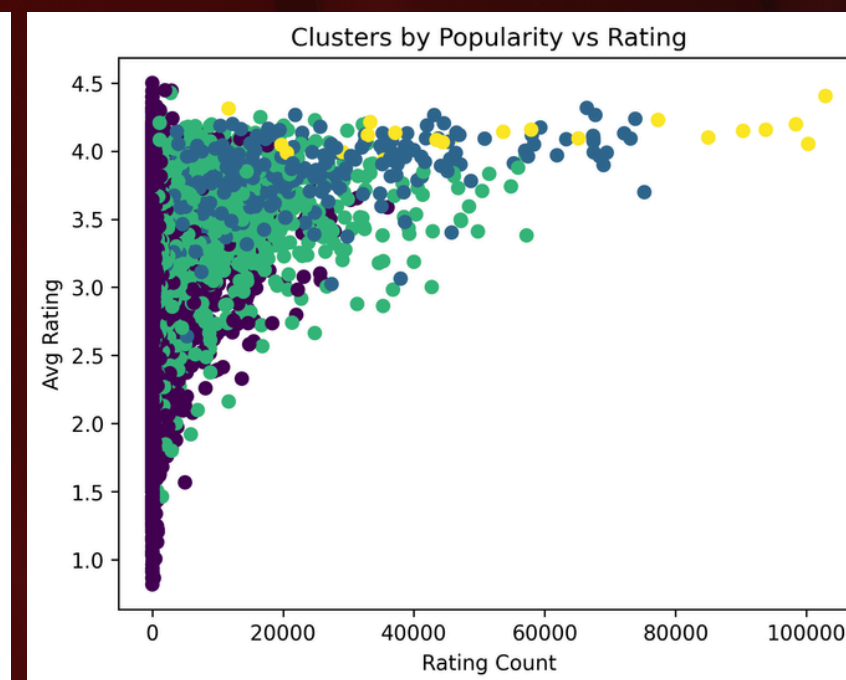
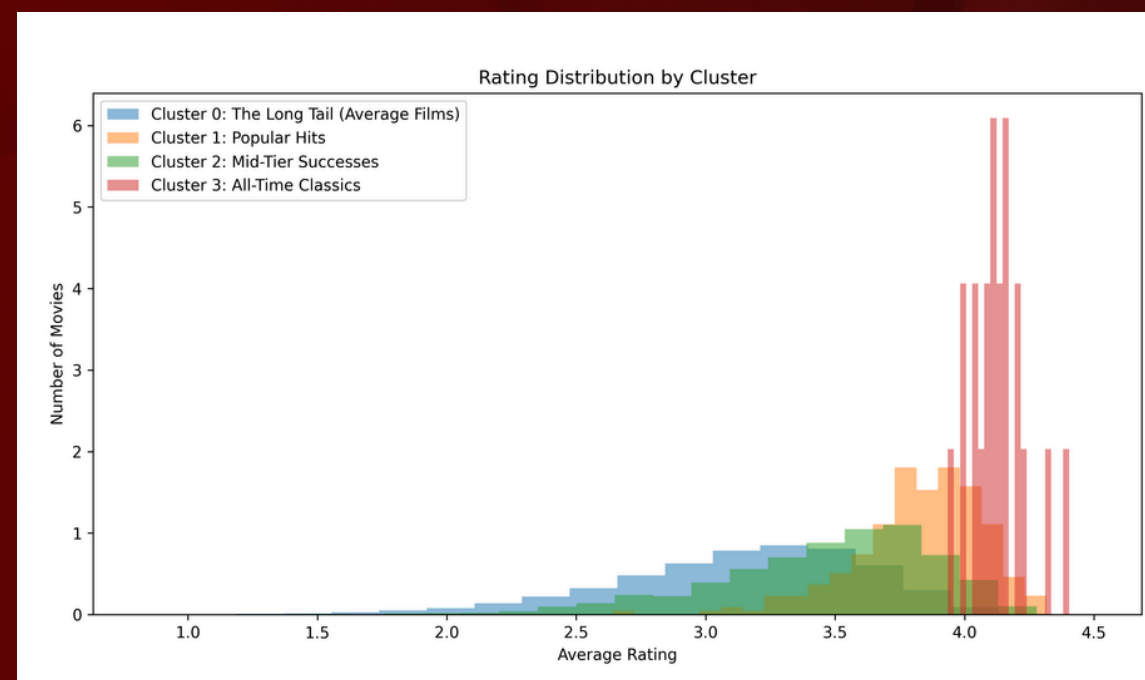
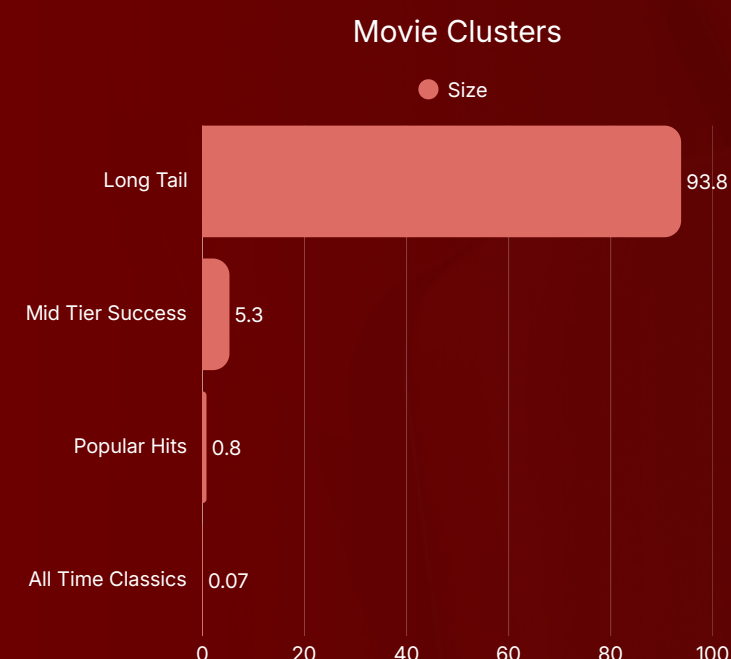
Understanding how users engage helps build better recommendation systems; not just what they watch, but how they rate and behave over time.



KEY FINDINGS

MOVIE HIERARCHY

K-Means revealed four movie tiers



Key Insight:

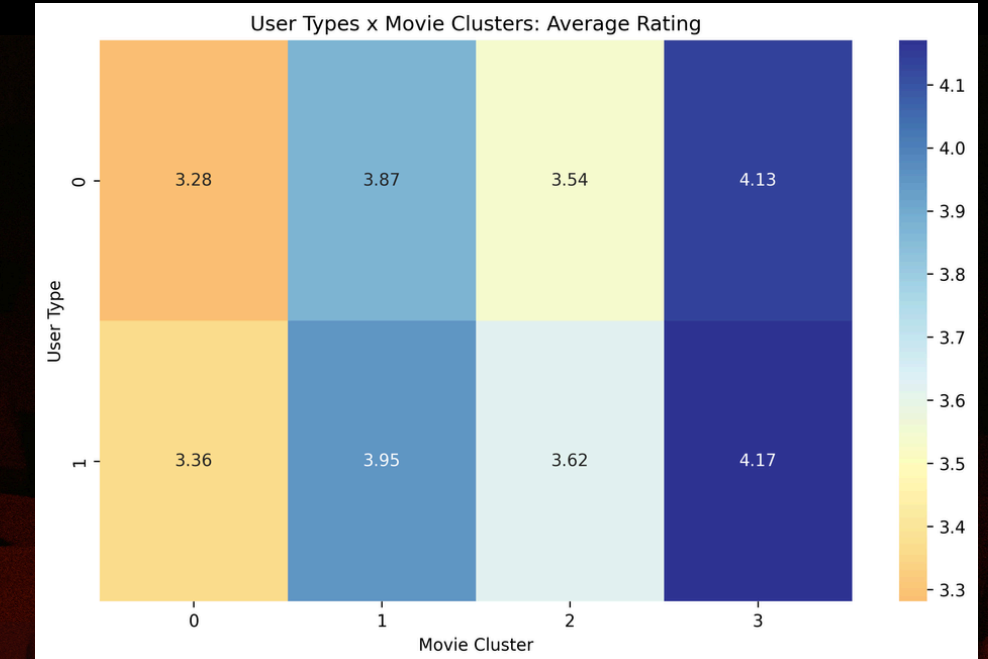
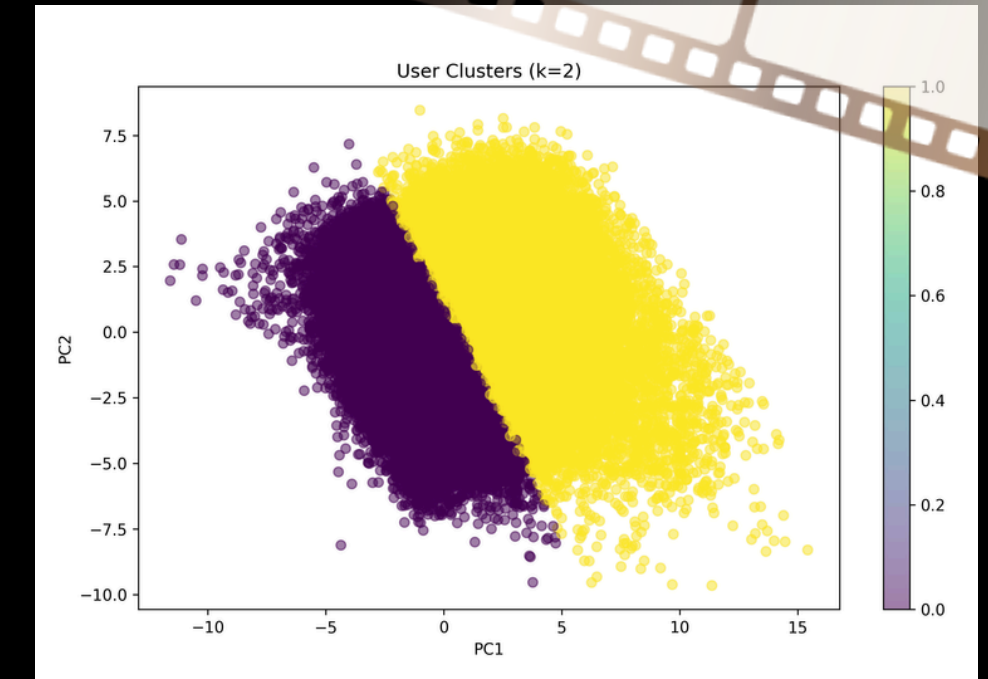
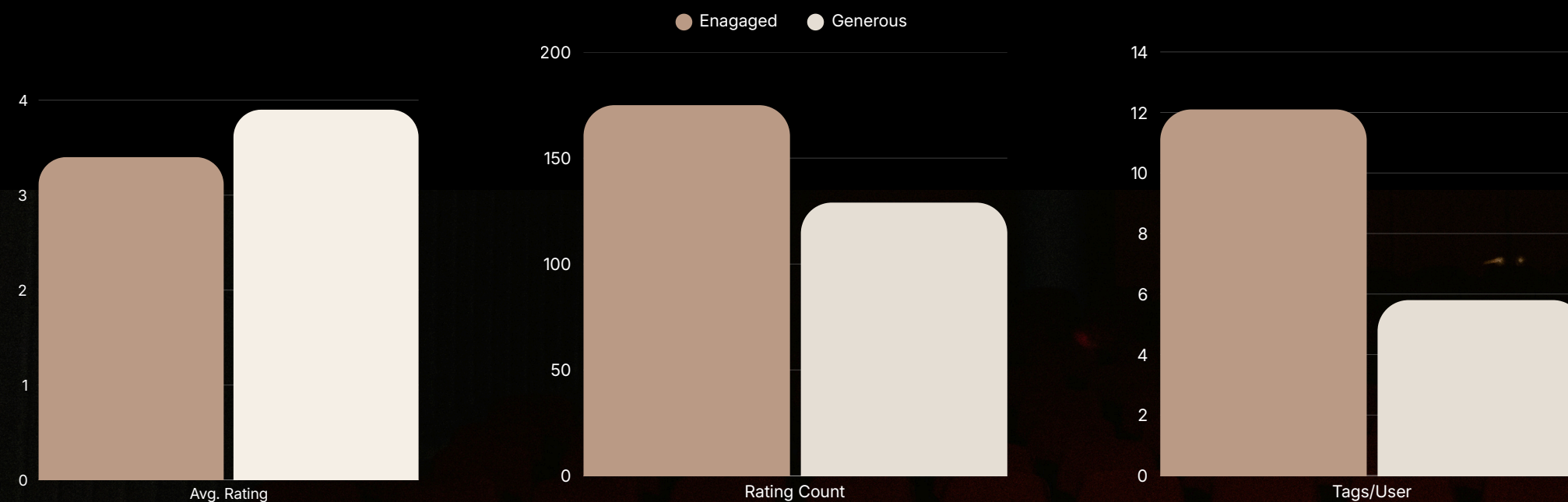
- Popularity (rating count) matters 18× more than rating quality (avg rating).
- A movie becomes a hit because people watch it, not because it's highly rated.
- Classics are the rare exception.



USER ARCHETYPES

K-Means revealed two user archetype:

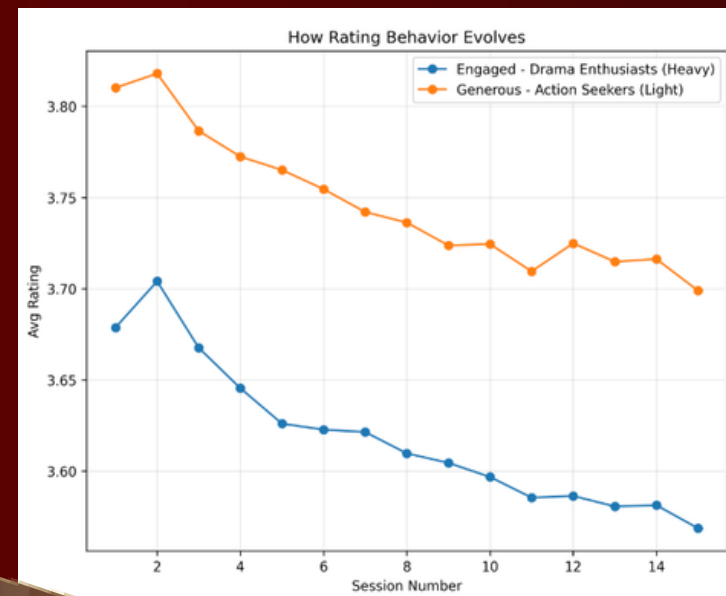
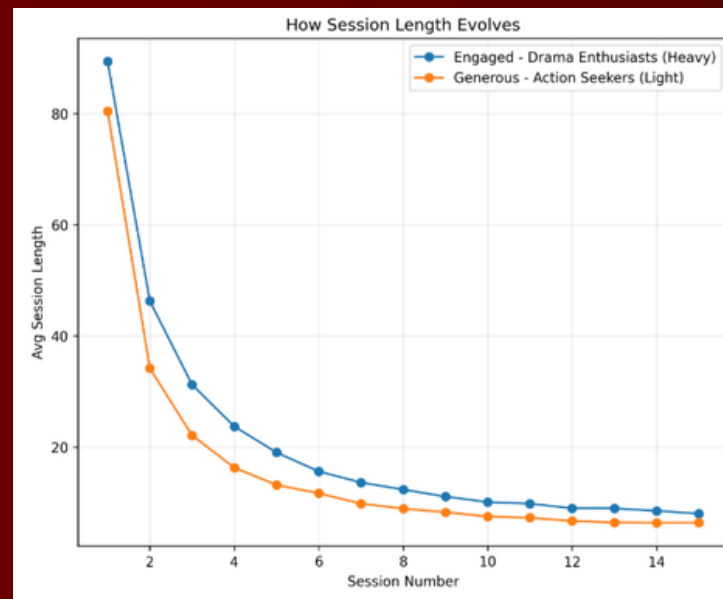
- Engaged Raters (65%) - More ratings, more tags, more critical
 - Prefer Drama
- Generous Raters (35%) - Fewer ratings, consistently higher scores
 - Prefer Adventure/Action



Key Insight:

- Users split by how they rate, not what they watch.
- Tagging is nearly independent of rating engagement.
 - A user can be a power rater without ever tagging, two separate dimensions of participation.

TEMPORAL PATTERNS



Burst Behavior: Users rate 11.5x more in their first session than later sessions.

- Engaged Raters: 89 → 7 ratings (steeper decline)
- Generous Raters: 80 → 8 ratings

Rating Recency: Engaged Raters watch older movies (14.5 years after release vs 13.9 years)

Consistency Over Time: Both groups rate lower over time (first-session inflation)

- Engaged: 3.68 → 3.46 (-0.22)
- Generous: 3.81 → 3.64 (-0.17)

Lifespan: Engaged Raters stay active 87 days longer (289 vs 202 days)

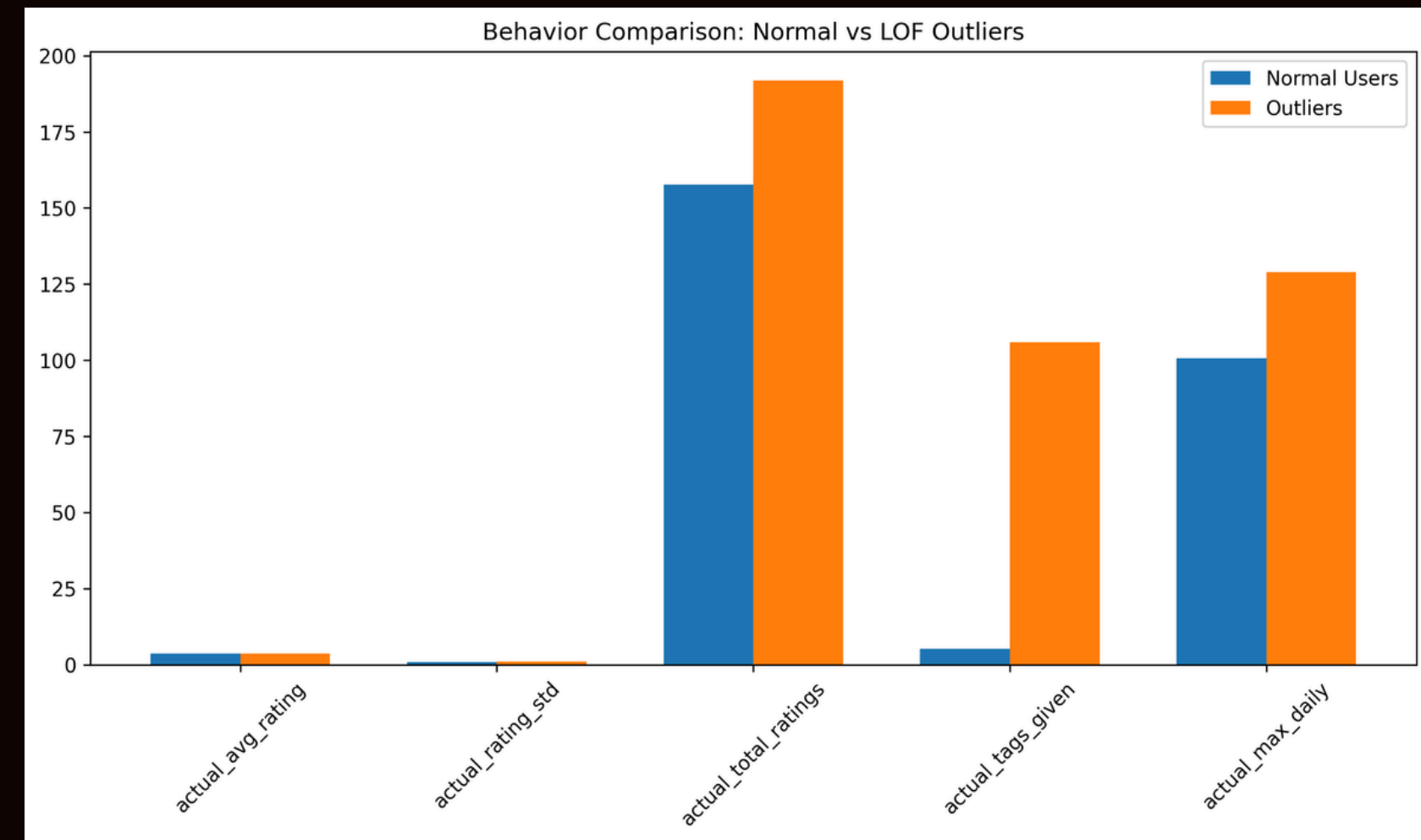
ANOMALIES

Local Outlier Factor (auto contamination):

- 4.7% of users are behavioral outliers
- Generous Raters have higher outlier rate (5.5% vs 4.3%)

What makes an outlier?

- 20x more tagging (106 vs 5 tags)
- Higher rating volume
- More inconsistent ratings
- More binge behavior



METHODOLOGY

Techniques Used

- K-Means Clustering to discover different groups (with PCA preprocessing to reduce noise)
- Decision Trees to validate the clusters and for interpretability
- Local Outlier Factor for anomaly detection
- Temporal Analysis to understand behavior over time (2-hour gap threshold)

Key Decisions

- Removed metadata richness from movie clustering - was dominating (100% importance), circular logic
- Used behavioral-only decision tree for users - genre preferences were overshadowing engagement patterns
- Depth = 4 for decision trees - interpretability over pure accuracy
- Auto contamination for LOF - let data determine outlier threshold naturally

VALIDATION & LIMITATIONS

Strengths:

- Movie decision tree: 95.5% accuracy
- Movie silhouette scores: 0.879
- Statistically significant temporal findings ($p < 0.001$)

Caveats:

- User silhouette score is modest. Behavior is a spectrum, not clean buckets
- MovieLens over-represents active, engaged users (not casual viewers)
- Newer movies have fewer ratings due to time, not quality
- K-Means assumes spherical clusters – may not be perfect for human behavior

CONCLUSION

Three key discoveries:

1. **Movie clusters are organized by engagement, not quality**
 - Long Tail to Classics, engagement matters 18x more than quality.
2. **Users are split by how they rate, not taste**
 - Generous vs Engaged. Rating quality > volume > tagging.
3. **Temporal patterns are universal**
 - Onboarding bursts, rating inflation, then decline.
 - Engaged users stay longer and explore deeper catalogs.

Main implication:

- Recommendation systems should model rating behavior, not just genre preferences.
- Two users who love the same genre can behave completely differently.



THANK YOU

Thoughts... questions?
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